Ocular Oncology and Pathology

Research Article

Ocul Oncol Pathol
DOI: 10.1159/000531006

Received: March 2, 2023
Accepted: May 3, 2023
Published online: May 11, 2023

Edge Creep: Increased Pigmentation at the Border of Choroidal Melanomas Treated with Plaque Brachytherapy

Elaine M. Binkley[a, b] Sean M. Rodriguez[b] Daniel E. Hyer[c] Connie J. Hinz[a] H. Culver Boldt[a, b]

[a]Institute for Vision Research, Department of Ophthalmology and Visual Sciences, University of Iowa, Iowa City, IA, USA; [b]Department of Ophthalmology and Visual Sciences, University of Iowa, Iowa City, IA, USA; [c]Department of Radiation Oncology, University of Iowa Hospitals and Clinics, Iowa City, IA, USA

Keywords
Choroidal melanoma · Uveal melanoma · Brachytherapy · Tumor regression

Abstract
Introduction: There is an increase in pigmentation that occurs in many tumors following plaque brachytherapy for choroidal melanoma. Correctly distinguishing between increased pigment at the tumor border versus true growth is imperative. We performed a retrospective review of patients treated with I-125 brachytherapy for choroidal melanoma at our institution to study this phenomenon. ***Methods:*** Records were reviewed for all patients undergoing plaque brachytherapy for uveal melanoma for a 5-year period (N = 195). Patients with iris and anterior tumors were excluded. Tumors treated more than 31 days after presentation were excluded. Fundus images for patients with increased pigmentation at any of the borders of the tumor at 6-month follow-up that extended beyond the initial pigmented margin were included (N = 20; 8 F, 12 M). Imaging at the last follow-up was reviewed, and it was confirmed that all tumors involuted appropriately with no evidence of local recurrence. The date of initial exam, time to treatment, and follow-up interval were recorded for each included patient. ***Results:*** Twenty patients (10%) exhibited increased pigment deposition at any of the borders of the tumor at 6-month follow-up that extended beyond the initial pigmented margin. Average tumor thickness was 3.2 mm (1.3–5.1); average largest tumor basal diameter was 11.6 mm (7–15.5). Average time from diagnosis to treatment was 25 days (17–31). Average length of follow-up was 35 months (16–68). No patient developed recurrence during the duration of follow-up, and 1 patient had developed metastasis. ***Conclusion:*** We describe the phenomenon of increased pigment deposition, "edge creep," at the borders of choroidal melanomas treated with plaque brachytherapy that gave the appearance of initial tumor growth but then subsequently remained stable over time. It is important that treating ocular oncologists be aware of this phenomenon to avoid unnecessary diagnosis of local recurrence.

Introduction

The Collaborative Ocular Melanoma Study (COMS) established plaque brachytherapy as an option for globe salvage in patients with medium-sized choroidal melanoma, and indications have subsequently been expanded

karger@karger.com
www.karger.com/oop

Correspondence to:
Elaine M. Binkley, elaine-binkley@uiowa.edu

to the treatment of small and large tumors [1–3]. The COMS defined local treatment failure as (1) initial height increase of 15% or 250-μm expansion of the tumor borders, and (2) an additional 15% increase in thickness or further 250-μm growth at any tumor border on a subsequent exam [2]. Clinically, ocular oncologists following treated tumors look for involutional changes such as decreasing thickness, increasing reflectivity, and decreasing vascularity on standardized echography, decreasing lipofuscin, and decreased subretinal fluid in addition to the absence of additional growth in order to ensure adequate tumor response after brachytherapy [4].

An important phenomenon that has been less well described but is important in assessing tumor response to treatment is the increase in pigmentation that occurs in many tumors following brachytherapy. This increase in pigmentation was described by Maheshwari and Finger (2018), who found an increase in tumor pigmentation in 64% of patients treated with palladium plaques [4]. A challenge arises when a previously amelanotic or lightly pigmented border of a tumor becomes more heavily pigmented following brachytherapy, and it can be unclear whether this represents increased pigment deposition in an area of tumor involution or treatment failure. This phenomenon has been known about since the time of the COMS, given the criteria for treatment failure after an additional 250-μm on a subsequent exam rather than the initial follow-up visit after treatment, but this has not been well described in the literature.

Given that local recurrence is associated with an increased risk for metastasis, correctly distinguishing between increased pigment at the tumor border which we will refer to as "edge creep" versus true growth is imperative. Here, we describe a series of patients with small and medium-sized melanomas treated with plaque brachytherapy who developed edge creep noted at the time of their first follow-up visit after brachytherapy but subsequently showed no further lateral expansion of pigment and had additional markers of tumor involution. We aim to make ocular oncologists aware of this phenomenon to avoid the assumption of treatment failure in these cases and prevent unnecessary re-treatment or enucleation.

Methods

This study was approved by the Institutional Review Board at the University of Iowa and adhered to the tenants of the Declaration of Helsinki. Records were reviewed for all patients undergoing plaque brachytherapy for uveal melanoma for a 5-year period from January 1, 2016 to December 31, 2020. Patients with iris and anterior tumors that could not be imaged with fundus photography were excluded. Patients whose tumors were treated more than 1 month following initial presentation were excluded due to the possibility of mild growth at the borders prior to treatment. Fundus images for each patient were reviewed, and patients for whom there was increased pigmentation at any of the borders of the tumor at their initial post-brachytherapy follow-up visit (around 6 months at our institution) that extended beyond the initial pigmented margin were included. Clock hours of creep and the number of quadrants involved were recorded. Patients for whom imaging was obtained on different cameras at presentation and initial follow-up, those who did not follow up at our institution after 6 months, and those with tumor biopsy that resulted in pigmentary changes at the biopsy site were excluded. Imaging at the last follow-up was reviewed, and it was confirmed that all tumors involuted appropriately with decreased thickness and no evidence of local recurrence. The date of initial exam, time to treatment, follow-up interval, and presence of metastasis were recorded for each patient. Optical coherence tomography (OCT) images for included patients were also reviewed.

Results

There were 195 patients treated with plaque brachytherapy during the study period. We identified 20 patients who met inclusion criteria. There were 8 female patients and 12 male patients. There were 3 right eyes and 17 left eyes. Average tumor thickness was 3.2 mm (1.3–5.1), and average largest tumor basal diameter was 11.6 mm (7–15.5). Average time from diagnosis to treatment was 25 days (17–31). Mean dose to the apex in included patients was 106 Gy (range: 92.9–116.8). All instances of edge creep were identified at the initial post-brachytherapy visit (slightly longer for some patients due to scheduling/patient transportation; average of 6.9 months; range: 6.1–8.6 months). Average length of follow-up was 35 months (16–68). Sixteen tumors were uniformly pigmented, 4 tumors were variably pigmented. The average number of clock hours with creep was 3.8 (range: 1–12), and the average number of quadrants with creep was 1.6 (range: 1–4). No patient developed recurrence during the duration of follow-up, and 1 patient had developed metastasis at the time of last follow-up (Table 1). Figures 1 and 2 show representative images at presentation, 6–7-month follow-up, and last follow-up for 2 of the included patients. Figure 3 shows representative images at presentation and 6-month follow-up for an additional patient with OCT images over the superior border of the tumor. OCT demonstrates that there is choroidal replacement present at the amelanotic portion of the tumor that became darker at the 6-month follow-up.

Table 1. Demographics, tumor characteristics, and follow-up

Number of eyes	20
Gender	8 female
	12 male
Eye involved	3 right
	17 left
Average tumor thickness, mm	3.2 (1.3–5.1)
Tumor largest basal diameter, mm	11.6 (7–15.5)
Tumor pigmentation	16 uniformly pigmented
	4 variably pigmented
Presence of metastasis at time of last follow-up	No: 19
	Yes: 1
Average time from diagnosis to treatment, days	25 (17–31)
Length of follow-up, months	35 (16–68)

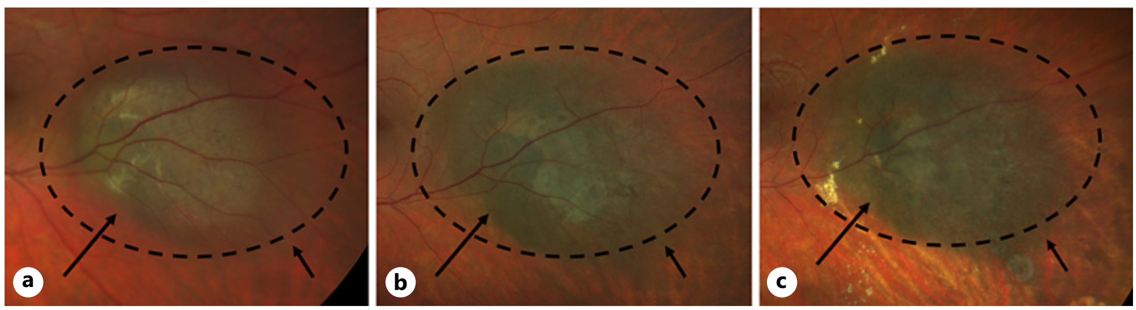

Fig. 1. Clarus fundus photograph for representative patient 1. Images at presentation (**a**), 7 months post-treatment (**b**), and 29 months post-treatment (**c**). Dashed circle shows the area around the tumor at presentation with an amelanotic surrounding rim and the same location on subsequent images for comparison. Arrows indicate areas of increased pigment that remained stable at the last follow-up.

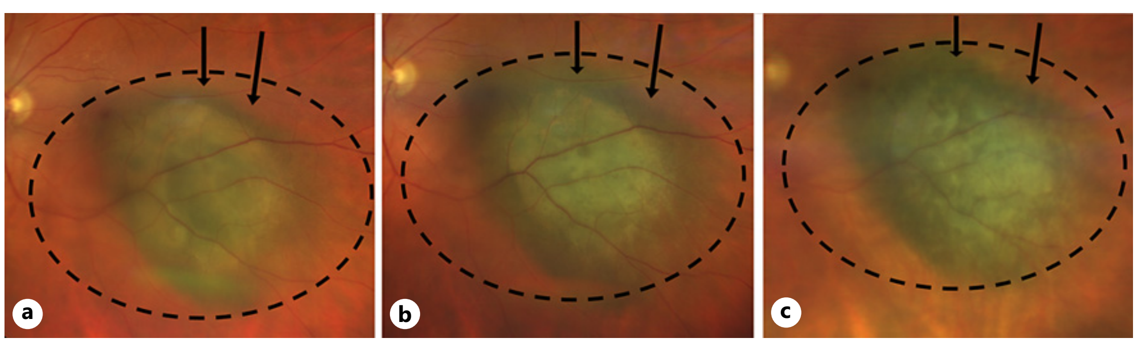

Fig. 2. Clarus fundus photograph for representative patient 2. Images at presentation (**a**), 6 months post-treatment (**b**), and 28 months post-treatment (**c**). Dashed circle shows the area around the tumor at presentation with an amelanotic surrounding rim and the same location on subsequent images for comparison. Arrows indicate areas of increased pigment that remained stable at the last follow-up.

Discussion

Given that local recurrence following plaque brachytherapy is associated with an increased rate of metastasis in patients with choroidal melanoma, it is imperative that recurrences be accurately detected and treated [5]. However, determining local recurrence is not always an easy task given the slow growth rate and variable regression patterns of these tumors [4, 6, 7]. An understanding of different tumor regression patterns following brachytherapy

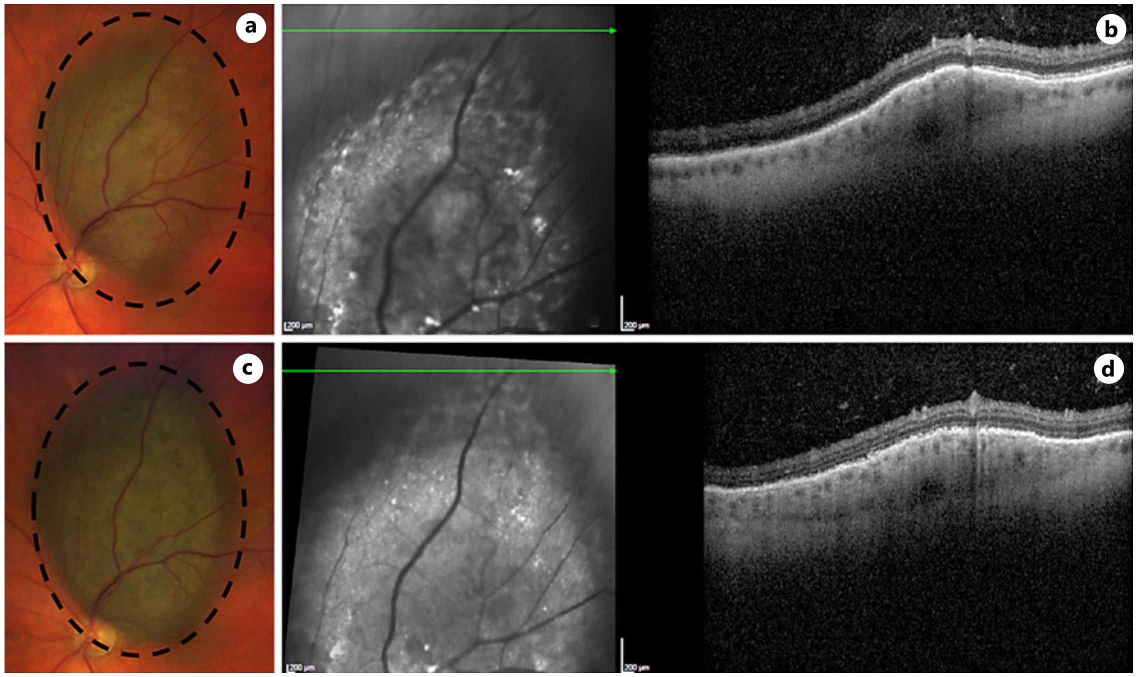

Fig. 3. Clarus fundus photograph for representative patient 3 with optical coherence tomography (OCT) over the tumor border. Clarus fundus photograph at presentation (**a**) and OCT over the superior tumor border (**b**) showing replacement of the choroid in the more lightly pigmented portion of the lesion. Clarus fundus photograph at 6-month follow-up (**c**) and OCT showing replacement of the choroid at the now more heavily pigmented border (**d**) (registration artifact shows that this is not in the exact same location as the OCT in b). Dashed circle shows increased pigment at all borders of the tumor.

is essential to ensuring that patients are either appropriately observed in the setting of regression or retreated/enucleated in the setting of recurrence. Here, we describe the phenomenon of increased pigment deposition, "edge creep," at the border of treated tumors following brachytherapy that subsequently remained stable over time in patients who did not develop local recurrence. This was observed in 10% of patients who were treated with brachytherapy within 1 month of presentation over a 5-year period at our institution, and we believe that it is important for ocular oncologists performing plaque brachytherapy to be aware of this regression pattern in order to avoid unnecessary diagnosis of local recurrence.

The concept of increased pigment deposition at the borders of treated tumors has been known since the time of the COMS, as demonstrated by the study criteria for recurrence specifying additional 250-µm growth at any tumor border after an initial increase in the borders. This 250-µm guideline suggests that there can be some growth/pigment deposition at the treated tumor border that does not represent true recurrence of the tumor. Increased tumor pigmentation is a known regression pattern, but this can be particularly challenging when it occurs at the border of a treated tumor, making it unclear whether there has been further growth [4]. A potential cause for observed increased pigmentation at the tumor borders is increased pigment deposition at the previously amelanotic edge of a tumor. Sometimes, especially in tumors with turbid subretinal fluid overlying the margin, the clinical border of the tumor can be more difficult to define. We also know that uveal melanomas can have microscopic extensions into the surrounding choroid [8]. It is possible that these cells become more pigmented when they are treated, and this may account for expansion of the margin. Another possible cause could be pigment-laden macrophages accumulating at the tumor border in response to radiation. It is possible that in some of our patients, this edge creep might represent tumor growth that occurred in the interval between baseline imaging and brachytherapy. Although the average time from baseline imaging to brachytherapy was 25 days, this might be long enough for rapidly growing tumors to enlarge. However, it seems faster than the growth rate for most uveal melanomas [6, 7]. While repeat imaging on the day of brachytherapy could confirm this potential cause, we did not image any of these eyes on the day of treatment.

Prior work has shown that more darkly pigmented tumors may regress more slowly, possibly due to mechanical effects of the increased melanin in the tumor cells [9]. It is possible that the baseline degree of pigmentation of the tumor may play a role in this phenomenon. This should be studied further with histopathologic analysis if any of these eyes require enucleation. Maheshwari and Finger showed that tumor pigment increased in 64% of cases in a series of 170 patients treated with palladium plaques [4]. Their manuscript reported the presence of pigmentary changes at 1, 3, 5, and 10 years and reported the overall change in pigmentation of the entire tumor rather than specifically at the borders. It is notable that all of the patients in our series had the increased pigment at the tumor borders detected at the patient's initial follow-up following brachytherapy (6 or 7 months) [4]. Given that many treatment failures occur later, our series suggests that the presence of early pigment is more suggestive of pigment deposition than true growth. At our institution, when edge creep is initially identified at the 6-month follow-up visit, we often see patients back at a shorter (3–4 month) interval to ensure stability prior to extending back to 6 months. We believe that this is a useful strategy, as a true treatment failure would be expected to have some continued growth in a 3–4 month time frame.

The isotype used for brachytherapy may also play a role in the pigmentary and retinal pigment epithelial changes that occur at the tumor border following brachytherapy. Regression patterns have been described following treatment with I-125, palladium, and ruthenium plaques [4, 9]. It is possible that the distance of the radiation source to the tumor as well as the tumor thickness and shape may play a role in the phenomenon of edge creep at the tumor borders. Studies using fundus autofluorescence have shown different patterns of retinal pigment epithelial atrophy at tumor borders, and this may also play a role in the appearance of increased pigment at the borders of treated tumors [10]. OCT imaging may also be useful in better understanding and documenting this phenomenon. Given the retrospective nature of this study, OCT images over tumor borders were variable and limited by the tumor location and registration artifacts. However, as in Figure 3, OCT over the tumor border can show choroidal replacement at the amelanotic border of tumors that may be harder to see clinically. Swept-source and widefield OCT may make this more practically useful in the future.

Our study is limited by the relatively small sample size and absence of enucleated eyes for histopathologic analysis. Further analysis of the specific tumor characteristics that are more likely to result in edge creep following brachytherapy should be performed in a larger series of patients.

Conclusion

We describe the phenomenon of early increased pigment deposition, edge creep, at the borders of small and medium-sized choroidal melanomas treated with plaque brachytherapy that gave the appearance of initial tumor growth but then subsequently remained stable over time. It is important that treating ocular oncologists be aware of this phenomenon to avoid unnecessary diagnosis of local recurrence. It is important that all tumor features including thickness, presence of decreasing subretinal fluid/exudative retinal detachment, regression of lipofuscin over the tumor surface, pigmentary changes, and basal dimensions be carefully monitored following treatment in order to ensure adequate tumor regression. The precise mechanism behind edge creep in treated choroidal melanomas warrants further histopathologic investigation and study in the future.

Statement of Ethics

This research adhered to the tenets of the Declaration of Helsinki. The study protocol was reviewed and approved by the Institutional Review Board at the University of Iowa (IRB #202210431). The Institutional Review Board at the University of Iowa deemed that written informed consent was not required for this study as it was a retrospective review.

Conflict of Interest Statement

The authors have no conflicts of interest to disclose.

Funding Sources

The authors have no funding sources to declare.

Author Contributions

Design of work and acquisition, analysis, and/or interpretation of data: E.M.B., H.C.B., C.J.H., and S.M.R. Drafting of the work/critical review of the manuscript and approval of the final version of the manuscript: E.M.B., H.C.B., D.E.H., C.J.H., and S.M.R.

Data Availability Statement

The data are available in the manuscript. Further inquiries can be addressed to the corresponding author.

References

1 Shields CL, Sioufi K, Srinivasan A, Di Nicola M, Masoomian B, Barna LE, et al. Visual outcome and millimeter incremental risk of metastasis in 1780 patients with small choroidal melanoma managed by plaque radiotherapy. JAMA Ophthalmol. 2018;136(12):1325–33.

2 Melia BM, Abramson DH, Albert DM, Boldt HC, Earle JD, Hanson WF, et al. Collaborative Ocular Melanoma Study (COMS) randomized trial of I-125 brachytherapy for medium choroidal melanoma. I. Visual acuity after 3 years COMS report no. 16. Ophthalmology. 2001;108(2):348–66.

3 Shields CL, Naseripour M, Cater J, Shields JA, Demirci H, Youseff A, et al. Plaque radiotherapy for large posterior uveal melanomas (> or =8-mm thick) in 354 consecutive patients. Ophthalmology. 2002;109(10):1838–49.

4 Maheshwari A, Finger PT. Regression patterns of choroidal melanoma: after palladium-103 ((103)Pd) plaque brachytherapy. Eur J Ophthalmol. 2018;28(6):722–30.

5 Ophthalmic Oncology Task Force. Local recurrence significantly increases the risk of metastatic uveal melanoma. Ophthalmology. 2016;123(1):86–91.

6 Char DH, Heilbron DC, Juster RP, Stone RD. Choroidal melanoma growth patterns. Br J Ophthalmol. 1983;67(9):575–8.

7 McLean IW, Foster WD, Zimmerman LE, Martin DG. Inferred natural history of uveal melanoma. Invest Ophthalmol Vis Sci. 1980;19(7):760–70.

8 Histopathologic characteristics of uveal melanomas in eyes enucleated from the Collaborative Ocular Melanoma Study. COMS report no. 6. Am J Ophthalmol. 1998;125(6):745–66.

9 Rashid M, Heikkonen J, Singh AD, Kivelä TT. Clinical predictors of regression of choroidal melanomas after brachytherapy: a growth curve model. Ophthalmology. 2018;125(5):747–54.

10 Navaratnam J, Bærland TP, Eide NA, Faber RT, Rekstad BL, Vavvas DG, et al. Fundus autofluorescence change as an early indicator of treatment effect of brachytherapy for choroidal melanomas. Ocul Oncol Pathol. 2019;5(6):445–53.